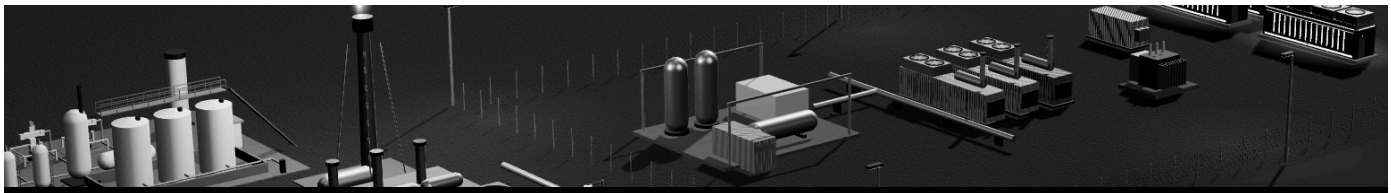
Cardinal Energy Buffalo batteries (Twp 64 Rge 16 W4), Athabasca County, Alberta: site sizing

What this gas could power, how much land it would take, and what it could earn each year.

2.6 MW	3	8 ac	194k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $583k	$8k to $16k	$8k to $599k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=GAS&FLOW=0.54

THE MATH

Methane in 0.73 MMcf/d of flared and vented gas, as pipeline-quality gas	0.54 MMcf/d
Energy in the gas	23.3 MMBtu per hour
Power at a heat rate of 9,000 Btu/kWh	about 2.6 MW
Compute (power less cooling and losses, /1.2)	about 2.2 MW
Compute modules (about 1 MW each)	3
Gas engines (running plus one spare)	3 at 2.5 MW
Land: 5 acres plus 1.2 acres per MW	about 8 acres
Gas bought per year (95% run time)	194k MMBtu
Gas payment, up to $3.00/MMBtu	up to $583k
Land lease at $1,000 to $2,000 per acre per year	$8k to $16k

Sized on the methane only: 0.73 MMcf/d (20.8 e3m3/d) at about 75% methane and 1,012 Btu per cubic foot, shown as 0.54 MMcf/d of pipeline-quality gas at 1,037 Btu. Heavier gases would add energy and are not counted. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Our own gas engines on a fenced parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Gas analysis (methane, H2S) at the batteries
- Gathering lines to bring the batteries' gas to one site
- Road and fiber access

Source: Petrinex public Alberta volumetric data, Oct 2025-Sep 2026 (read Nov 12, 2026): the four batteries flared and vented 20.8 e3m3/d (0.73 MMcf/d) on average in Jul-Sep 2026 (20.5 e3m3/d over Oct 2025-Sep 2026). Methane share of 75% is an estimate; the sizing counts the methane only. This is an estimate from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.